

Trackformer 1.2

Architecture and Forecast-State Evolution

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<https://github.com/yu314-coder/typhoon-predict>

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Abstract

Trackformer 1.2 advances an atmospheric basin state and a translating pressure core to forecast tropical-cyclone motion and central pressure. Recurrent convolutional networks evolve the fields; multiscale attention supplies environmental context. The reported centre is associated with a depression in the predicted core, and central pressure is sampled from that same field. Twenty six-hour transitions produce a five-day forecast. This paper describes only the released architecture, input contract, update equations, objectives and ensemble mechanism. It contains no benchmark comparisons or skill claims.

1 Design and forecast state

The system is not a route predictor followed by an independently drawn pressure image. It evolves pressure and atmospheric states, proposes a translation of a storm-following frame, then associates a centre with the predicted pressure within that frame. Translation is learned and steering-conditioned: this is neither a pressure-only tracker nor a complete atmospheric numerical solver.

At step k , the recurrent state is

$$s_k = (g_k, h_k^g, c_k, h_k^c, a_k, o_k, z_k, m_k, v_k; G, R, I, M).$$

Here g is the normalized eight-channel basin field; h^g its memory; c the core pressure; h^c core memory; a its anomaly relative to the basin; o the core-grid origin; z the associated storm centre; m displacement per six hours; and v auxiliary maximum wind. Static geography G , regional coordinates R , issue intensity I and masks M stay fixed.

Stage	Function
Initialization	Past analyses and issue observations initialize fields and memories.
Attention	Multiscale environmental tokens condition basin memory.
Basin transition	Steering, backward transport and learned tendencies evolve eight fields.
Core transition	A translated frame carries a recurrent pressure anomaly.
Readout	Field-based association yields route and central pressure; pressure is reconstructed geographically.

2 Inputs, geometry and outputs

Nine past analyses at $-48, -42, \dots, 0$ h form a tensor of shape $B \times 9 \times 8 \times 25 \times 33$. The basin spans 60 to 0°N and 100 to 180°E at 2.5° spacing. Channels are

$$[\text{MSLP}, Z_{500}, u_{850}, v_{850}, u_{500}, v_{500}, u_{200}, v_{200}],$$

in hPa, metres and metres per second. Channel normalization is $\tilde{x}_j = (x_j - \mu_j)/\sigma_j$, with the released training-only constants. Statistics are not refitted for each forecast.

Optional regional pressure history has shape $B \times 9 \times 1 \times 121 \times 121$ on an issue-anchored patch sampled every 0.25° , with offsets of $\pm 15^\circ$. A detail-availability flag chooses native regional history or coarse basin pressure during initialization. The regional coordinate grid is still supplied when native observations are unavailable.

Four static channels encode latitude/90, longitude/180 $- 1$, land fraction and elevation/8000 m. The issue state supplies latitude/longitude, east/north motion in km per six hours, maximum wind in knots, central pressure in hPa and two intensity-availability masks. Only observations and analyses at or before issue time may enter inference; retrospective data availability is a separate operational question.

The internal core is a 65×65 grid at 20-km sampling over ± 640 km. Its field is a learned reconstruction, not native 20-km observations. For each of twenty leads (6, 12, $\dots$, 120 h), outputs include basin fields, regional pressure, core pressure/coordinates, storm centre, central pressure, domain masks and auxiliary maximum wind. This implementation does not output quadrant wind radii or a resolved surface-wind map.

3 Neural modules

The released non-tiny architecture has 21,452,595 trainable parameters.

3.1 Multiresolution evolution networks

The basin network receives 44 channels: eight fields, 32 memory features and four static channels. Encoder widths are 72, 144, 288, 432; residual-block depths are 2, 2, 4, 4. A 3×3 stem and stride-two convolutions produce four scales. The decoder uses 1×1 projections, bilinear upsampling, additive skip connections and one residual block at each reconstructed scale. Output heads emit ten transition channels and 32 recurrent features. Eight transition channels are field tendencies and two are flow corrections.

Each residual block comprises a depthwise 5×5 convolution, eight-group normalization, 1×1 expansion by four, GELU and 1×1 contraction. A learned residual multiplier starts at 0.01. Transition heads start at zero. The core network uses widths 64, 128, 256, 384 and the same depths. Its 45 inputs are core pressure, eight environmental fields, 32 core-memory channels and four geographic channels. Three outputs represent one anomaly tendency and two local-flow corrections.

3.2 Convolutional recurrent memory

Basin and core memories each have 32 channels. History encoders receive 72 flattened basin channels or nine pressure-history channels, respectively. Given advected memory h and new evolution features f ,

$$\begin{aligned}(r, u) &= \text{sigmoid}(W_g * [h, f]), \\ q &= \tanh(W_q * [r \odot h, f]), \\ h' &= (1 - u) \odot h + u \odot q.\end{aligned}$$

The gate convolution emits 64 channels split into reset and update gates; the candidate convolution emits 32. Forecast leads share these recurrent modules rather than using twenty independent predictors.

3.3 Environmental attention

A 1×1 convolution projects $[g, h^g, G]$ from 44 to 64 channels. Average pooling at sizes/strides three and five with ceiling mode yields $9 \times 11 + 5 \times 7 = 134$ environmental tokens. Two pre-normalization Transformer layers use width 64, four heads, feed-forward width 128, ReLU activation and zero dropout. Each of the 825 basin cells supplies a query projected from its memory to width 64. Four-head cross-attention retrieves context; a zero-initialized $64 \rightarrow 32$ projection adds it to basin memory before the transition. No separate positional embedding is added: geographic coordinates are already among the token inputs.

4 Observation-conditioned initialization

Write $\mathcal{S}(F, \xi)$ for bilinear field sampling. Sampling uses FP32 geometry, aligned corners and border padding for the basin; transported core anomalies use zero padding. All nine pressure histories are sampled onto the core grid centred at the issue observation. A detail mask chooses native pressure or coarse pressure. The latest map initializes c_0 , while all nine maps initialize the core memory.

If issue central pressure is valid, the initial core is corrected by

$$c_0(x, y) \leftarrow c_0(x, y) + (\tilde{p}_{\text{issue}} - c_0(0, 0))e^{-(x^2+y^2)/(2 \cdot 100^2)}T(x, y),$$

where x, y are kilometres and $T = \text{clip}((640 - \sqrt{x^2 + y^2})/160, 0, 1)$. The anomaly is $a_0 = c_0 - \mathcal{S}(g_{0, \text{MSLP}}, \xi_0)$. This compact prior is applied only at initialization. Future scalar pressure is not repeatedly inserted into the field.

5 One six-hour transition

5.1 Basin transport

Learned softmax weights blend physical winds at 850, 500 and 200 hPa. Their initialization is $(0.4, 0.4, 0.2)$, not a fixed inference constraint. For network output d ,

$$U_k = \sum_{\ell} \alpha_{\ell}(u_{\ell}, v_{\ell})_k + 15 \tanh(d_{8:10}) \text{ m s}^{-1}.$$

Backtracing constructs departure coordinates $D(U_k)$ and

$$g_{k+1} = \mathcal{S}(g_k, D(U_k)) + 0.2 \tanh(d_{0:8}).$$

Six hours convert 1 m/s to 21.6 km. Coordinate conversion uses 111.2 km per latitude degree and $111.2 \max(\cos \phi, 0.2)$ km per longitude degree. The attention-conditioned memory is advected by the same departure grid and updated with new features. Tendencies are in normalized channel units. This semi-Lagrangian-style update is not a complete or exactly conservative atmospheric equation solver.

5.2 Moving-frame proposal

A $44 \rightarrow 64 \rightarrow 3$ GELU MLP reads sampled 32-channel memory, eight basin fields, previous motion divided by 100, and two issue-intensity masks. For output b ,

$$m_{k+1} = 0.7(21.6)U_k(z_k) + 0.3m_k + 100 \tanh(b_{0.2}).$$

This displacement, in km per six hours, defines a new origin relative to z_k . It proposes where the pressure core moves, but is not the final reported centre. The third output updates auxiliary wind: $v_{k+1} = \max(0, v_k + 5b_2)$.

5.3 Relative core transport

The core network reads the previous pressure core, its old sampled environment, core memory and geography at the proposed origin. Local velocity is $U^{\text{local}} = \mathcal{S}(U_k) + 5 \tanh(d_{1.3}^c)$. Subtracting $m_{k+1}/21.6$ removes bulk frame translation. Backward coordinates also account for the offset between the previous associated centre and its previous grid origin. On this relative departure grid D_c ,

$$\begin{aligned} a_{k+1} &= \mathcal{S}(a_k, D_c) + 0.1d_0^c, \\ c_{k+1} &= \mathcal{S}(g_{k+1, \text{MSLP}}, \xi_{k+1}) + Ta_{k+1}. \end{aligned}$$

Unlike the basin tendency, d_0^c is not bounded by $\tanh$. Core memory uses the same relative sampling followed by its gated update.

5.4 Centre association and central pressure

Convert core pressure to hPa: $P = \sigma_p c_{k+1} + \mu_p$. For cells within 300 km of the proposed origin,

$$w_i = \text{softmax}_i \left(-P_i/2 - r_i^2/(2 \cdot 180^2) \right).$$

Cells beyond 300 km are excluded. The weighted mean of cell latitude/longitude is z_{k+1} . Central pressure is the bilinear sample $\mathcal{S}(P, z_{k+1})$, not necessarily the grid minimum. The spatial prior associates the forecast with this storm rather than an unrelated remote depression.

5.5 Geographic pressure output

The issue-anchored regional output is

$$\hat{p}_R = \mathcal{S}(g_{k+1, \text{MSLP}}, R) + \mathcal{S}(Ta_{k+1}, R; \text{zero padding}),$$

then de-normalized to hPa. This basin-plus-core reconstruction is generated by the model itself; future observed fields and display-only vortices are not substituted. Its 0.25° sampling is not proof of native resolution. Masks indicate valid basin coverage. Resampling and localized association mean the reported centre, central pressure and regional map minimum need not coincide exactly. A display must preserve those distinctions rather than relocate fields.

6 Autoregression and learning objectives

Inference initializes once and repeats the transition twenty times, using only fixed issue context and its own predicted future state. The implemented single-step loss is

$$L = L_{\text{env}} + L_{\text{regional}} + 0.2L_{\text{route}} + 0.5L_{\text{pressure}} + 0.05L_{\text{wind}} \\ + 0.25L_{\Delta p} + 0.1L_{\Delta z}.$$

The final terms require valid preceding-target labels. Environment loss is cosine-latitude-weighted Smooth L1 in normalized units. Regional loss includes Smooth L1 and 0.2 times adjacent-grid gradient errors on each axis. Absolute regional-field supervision excludes the inner 150 km around the target centre, because reanalysis can underresolve observed central pressure. This is a supervision choice, not proof of correct inner-core structure.

Route residuals use a 100-km scale; pressure and wind use 20 hPa and 30 kt. Pressure-change error uses 10 hPa and displacement-change error uses 100 km. All objectives use explicit masks. They train field evolution and readout together, without adding an independent central-pressure output head. Training-only normalization, storm-level split boundaries and causal input selection remain essential implementation constraints.

7 Single forecasts and 50-member means

Evaluation-mode inference is deterministic for fixed inputs and weights. The documented ensemble perturbs only normalized historical weather: independent Gaussian fields are spatially smoothed with kernels five (basin) and nine (regional), centred, unit-scaled per map and multiplied by amplitudes 0.025 and 0.015. Issue centre, motion, intensity, masks and geography remain fixed. Each member performs a complete recurrent rollout.

With $M = 50$, routes and central pressures are averaged separately: $\bar{z} = M^{-1} \sum_m z^{(m)}$ and $\bar{p} = M^{-1} \sum_m p^{(m)}$. Physical pressure fields are averaged on common geographic grids, not by averaging unaligned moving-core array indices. In general,

$$\bar{z} \neq \arg \min_x \bar{P}(x), \quad \bar{p} \neq \min_x \bar{P}(x).$$

These are perturbed-input realizations of one model, not 50 trained networks or learned latent samples. Spread is not automatically calibrated uncertainty. Member counts, perturbation settings and seed policy must accompany the outputs.

8 Source identity and use boundary

The public name is **Trackformer 1.2**. Its manifest retains the internal checkpoint identity, 1.2.73 epoch 4, and exact weight/source hashes. `model.py` implements environmental attention and added temporal losses; `baseline_model.py` implements moving-core evolution and readout; `v165.base.py` supplies sampling, evolution blocks and memories. All are in `models/trackformer_1.2_field/`.

This paper describes mechanisms only. Learned reconstructions, limited domain, input-source differences and retrospective availability constrain interpretation. The model remains a research artifact, not an operational warning system.